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324811(24)

B. E. (Eighth Semester) EXAMINATION, Nov.-Dec., 2014

(Old Scheme)

(Branch : Electrical)

MODERN CONTROL SYSTEM

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : Assume suitable data wherever necessary. Section (a) is compulsory in all questions. Solve any *two* sections out of (b), (c), and (d) in each question.

1. (a) Define limit cycle. 2

   (b) Explain stability analysis using describing functions. 7

   (c) Discuss second method of Liapunov with four stability theorems. 7

   (d) Consider a nonlinear system described by the equations : 7

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\left(1 - |x_1|\right)x_2 - x_1$$

Find the region in the state-plane for which the equilibrium state of the system is asymptotically stable.

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2. (a) Write at least two advantages of State Space Analysis. 2

   (b) Consider a control system with state model : 7

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix}[u]; \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

u = unit step*.

Compute the state transition matrix and therefrom find the state response, i. e., $x(t), t > 0$.

   (c) Show that if there is poles-zero cancellation in the input-output transfer function of a linear time-invariant discrete-time system, the system will be either uncontrollable or unobservable. 7

   (d) Explain state equations in CCF, OCF and Diagonal Canonical form with one suitable example. 7

3. (a) What is Servo system ? 2

   (b) Explain Ackermann's formula for Pole Placement in detail. 7

   (c) Discuss Pole placement design with example. 7

   (d) Write a short note on design of full and reduced order state observers with suitable example. 7

4. (a) What is discrete system ? 2

   (b) A discrete-time system has the transfer function : 7

$$T(z) = \left[4z^3 - 12z^2 + 13z - 7\right] / \left[(z-1)^2 (z-2)\right]$$

Determine the state model of the system in :

   (i) Phase variable form

   (ii) Jordon canonical form

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(c) Discuss Stability Analysis using Bilinear transformation Method and Jury's stability test. 7

(d) Explain reconstruction original signals from sampled signals with mathematical proof. 7

5. (a) Define optimal control system. 2

(b) Explain solution of the Discrete Riccati Equation by Recursive method of solving the Riccati equation with suitable mathematical example. 7

(c) Explain solution of the discrete Riccati Equation by Eigen value-Eigen vector method of solving the Riccati equation. 7

(d) Write a short note on Discrete Maximum (Minimum) Principle. 7

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